

GREEN HOUSE MONITORING BASED ON IOT

A Project Report

*Submitted in partial fulfilment of the requirements for the
award of the degree of*

BACHELOR OF TECHNOLOGY

In

ELECTRONICS AND COMMUNICATION ENGINEERING

By

M.SOMA SEKHAR (20B91A04E9)

M.SAI DEEPAK (20B91A04F2)

R.SIDDARDHA (20B91A04K4)

P.PRAHARSHA LAL (20B91A04I3)

Under the esteemed guidance of

SRI.Y.S.S SRIRAMAM, M.Tech

Assistant Professor, ECE Department



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

S.R.K.R. ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTU, KAKINADA)

(Recognized by A.I.C.T.E., Accredited by N.B.A & N.A.A.C with 'A⁺' Grade, New Delhi)

CHINNA AMIRAM, BHIMAVARAM

(2020-2024)

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(Recognized by A.I.C.T.E., Accredited by N.B.A & N.A.A.C with 'A⁺' Grade, NewDelhi)

CHINNA AMIRAM, BHIMAVARAM-534204

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING



BONAFIDE CERTIFICATE

This is to certify that this project work entitled

“GREEN HOUSE MONITORING BASED ON IOT ”

is the bonafied work of

**Mr. M.SOMA SEKHAR (20B91A04E9), Mr. M.SAI DEEPAK (20B91A04F2),
Mr. R.SIDDARDHA (20B91A04K4), Mr P.PRAHARSHA LAL(20B91A04I3) of the
final year B. Tech submitted in partial fulfillment of requirements for the award of
Degree in Bachelor of Technology in Electronics and Communication Engineering dur-
ing the academic year 2020-2024.**

Guide:

Mr. Y.S.S.SRIRAMAM

M.Tech, Assistant professor

Department of ECE

Head of the Department:

Dr. N. UDAYA KUMAR

M.Tech, Ph.D, M.I.S.T.E, S.M.I.E.E.E, F.I.E.T.E, F.I.E

Department of ECE

CERTIFICATE OF EXAMINATION

This is to certify that we have examined the report and here by accord our approval of it is a project carried out and presented in a manner required for its acceptance on a partial fulfilment for the award of the degree of **BACHELOR OF TECHNOLOGY** in **ELECTRONICS AND COMMUNICATION ENGINEERING** for which it had been submitted. This approval does not necessarily endorse or accepts every statement made, opinion expressed or conclusions drawn as recorded in the project report, it only signifies the acceptance of the report for the purpose for which it is submitted.

EXTERNAL EXAMINER

INTERNAL EXAMINER

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PROJECT ASSOCIATES

M.SOMA SEKHAR
M.SAI DEEPAK
R.SIDDARDHA
P.PRAHARSHA LAL

DECLARATION

This is to certify that the project entitled “**GREEN HOUSE MONITORING BASED ON IOT**” which is submitted by **M.SOMA SEKHAR(20B91A04E9), M.SAI DEEPAK(20B91A04F2), R.SIDDARDHA (20B91A04K4), P.PRAHARSHA LAL (20B91A04I3)**, partial fulfilment of the requirement for the award in degree in B.Tech in Electronics and Communication Engineering of S.R.K.R Engineering college, affiliated to JNTU KAKINADA. It comprises only our original work and due acknowledgment has been made in text to all other material used.

Date:

M.SOMA SEKHAR
M.SAI DEEPAK
R.SIDDARDHA
P.PRAHARSHA LAL

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ABSTRACT

This project is based on human less maintenance of agriculture . In this the moisture of the soil, temperature and light intensity of the place will maintained automatically. Although we can predict the upcoming weather, there can be changes like sudden rain or harsh sunlight. These unpredicted changes in weather can destroy the crops. To avoid this, this system will automatically maintain the conditions which are favourable for the crops. If there is sudden rain and if the temperature is too low for the crops, the system will turn on the heat lamp due to which the temperature will be maintained. If due to high temperature, the moisture of the soils gets reduced, the system will turn on the water pump to maintain the moisture. Some crops need specific amount of light. If there is a situation that the surrounding light is low for the crops, then the system will turn on the light bulb to maintain the light intensity. In all these situations the farmer need not to run behind things like turning on water pump, light bulb, fan, etc. This system will take care of the maintenance of agriculture. All this data will be transmitted to the mobile device of the farmer. A smart greenhouse system utilizing IoT and 8051 microcontrollers optimizes plant growth through continuous monitoring. Components like temperature and soil moisture sensors, LEDs, and motors are integrated for efficient control. The system regulates environmental factors like temperature, humidity, light, soil moisture, and CO2 levels ensuring optimal conditions. Growers receive real-time insights and alerts, enhancing decision-making for improved yield and resource efficiency. An H-bridge circuit and water pump enable precise control of motor direction and speed. LCD display provides convenient monitoring, while LEDs enhance light control within the greenhouse. The system reduces resource wastage, environmental impact, and supports remote monitoring. This project illustrates the benefits of IoT in agriculture and addresses challenges with suggestions for future enhancement.

CHAPTER 1

INTRODUCTION

1.1 Project Objective:

The objective of this project is to design and implement an Internet of Things (IoT) based greenhouse monitoring system aimed at optimizing plant growth conditions while minimizing resource consumption and environmental impact. The system will utilize various sensors to continuously monitor crucial environmental parameters such as temperature, humidity, soil moisture, light intensity, and CO₂ levels within the greenhouse. By collecting real-time data, the system will provide insights into the greenhouse environment, enabling remote monitoring and control through a user-friendly interface. The overarching goal is to create a sustainable and efficient solution that empowers growers to make informed decisions, improve crop yield, conserve resources, and reduce greenhouse gas emissions in agricultural practices. When sudden changes in weather can have detrimental effects on agriculture, impacting crop growth and yield. Unpredicted rainfall may lead to crop destruction, while insufficient rainfall can result in crop drying. Similarly, high temperatures can be detrimental to crops, and certain crops require specific light conditions for optimal growth. Insufficient light can hinder proper crop development. To address these challenges, farmers often need to manage various factors such as water supply, lighting, ventilation, and more simultaneously.

1.2 Project Motivation:

To alleviate the burden on farmers, a system has been developed to automate these tasks and provide real-time data to farmers via their mobile devices. This system helps reduce the need for manual labour and enables farmers to manage large-scale plantations more efficiently with fewer workers. One critical component of this system is the soil moisture sensor, which detects moisture levels in the soil. The sensor's resistance changes based on soil moisture: lower resistance indicates higher moisture levels, while higher resistance indicates lower moisture levels.

When farmers are away from their farms, they often worry about maintaining proper

conditions for their plants, especially regarding water supply. This automated system takes care of watering plants based on soil moisture levels and keeps farmers informed by sending alerts and updates to their cell phones. The system operates independently without requiring constant manual intervention, thanks to the control provided by Arduino and communication facilitated by GSM modules for sending alert messages this Completely Automated System for agriculture not only streamlines farm management tasks but also ensures optimal conditions for crop growth, reduces the reliance on manpower, and provides peace of mind to farmers even when they are not physically present on the farm.

1.3 Problem Statement:

In traditional greenhouse agriculture, monitoring and controlling environmental parameters crucial for optimal plant growth are often labor-intensive, inefficient, and prone to errors. The lack of real-time data and automated systems leads to suboptimal conditions, resulting in reduced crop yields, wasted resources, and increased environmental impact. Additionally, with the growing concern for sustainable agriculture practices, there is a pressing need for solutions that minimize energy consumption and greenhouse gas emissions.

To address these challenges, there is a demand for an Internet of Things (IoT) based greenhouse monitoring system that can continuously collect data on key environmental variables such as temperature, humidity, soil moisture, light intensity, and CO₂ levels. Such a system should provide growers with real-time insights into the greenhouse environment, allowing them to remotely monitor and control conditions to optimize plant growth while conserving resources and reducing environmental footprint. Thus, the problem at hand is to develop an efficient and scalable IoT solution that enables precise monitoring and control of greenhouse conditions, ultimately leading to improved crop yields, resource conservation, and sustainability in agricultural practices.

CHAPTER 2

LITERATURE SURVEY

In this section, Literature survey of Greenhouse Monitoring System Using IOT is presented.

2.1. Introduction:

Greenhouses play a crucial role in creating controlled environments suitable for optimal plant growth. However, traditional manual watering methods can be time-consuming for farmers, requiring constant attention and effort. To address this issue, modern greenhouses are integrating automated systems such as plant sprinklers, significantly reducing the time spent on watering plants compared to manual methods. This automation not only saves time but also helps in conserving water, as it ensures that plants receive the precise amount of water they require, reducing wastage due to overwatering or underwatering caused by human error. With the widespread development and accessibility of the internet across the globe, technology has revolutionized daily activities. The Internet of Things (IoT) has emerged as a powerful tool, enabling devices to connect and communicate seamlessly. In the context of agriculture, IoT technology has facilitated the development of advanced systems like the fertigation control device discussed here, which leverages sensors and the ESP8266 module for connectivity.

The hardware design of this system incorporates a micro controller for control purposes. The main objective of this tool is to automate the watering process based on soil moisture levels, ensuring optimal conditions for plant growth. It provides real-time monitoring of soil moisture levels, displaying the information on an LCD screen and through the Blynk application, which can be accessed from anywhere with internet connectivity.

The primary advantage of such a system is its ability to minimize human intervention in agricultural maintenance tasks. By automatically adjusting watering schedules based on soil moisture levels, the system ensures that plants receive the right amount of water at the right time, leading to improved crop health and yield. Moreover, the system can respond

dynamically to unforeseen changes in weather conditions, such as sudden rain or extreme sunlight, which can adversely affect crops. For instance, it can activate heat lamps during cold spells or turn on the water pump to maintain soil moisture during periods of high temperature. In essence, this system not only streamlines agricultural operations but also enhances productivity by creating and maintaining optimal growing conditions for crops. By transmitting crucial data and alerts to farmers' mobile devices, it empowers them to monitor and manage their crops efficiently, contributing to sustainable and efficient agricultural practice.

2.2. Research Paper:

I)BY USING ARDUINO UNO BOARD:

- https://www.researchgate.net/publication/347900320_Greenhouse_Monitoring_System_Using_IOT
- This project is based on human less maintenance of agriculture . In this the moisture of the soil of the place will maintained automatically. Although we can predict the upcoming weather, there can be changes like sudden rain or harsh sunlight. These unpredicted changes in weather can destroy the crops. To avoid this, this system will automatically maintain the conditions which are favorable for the crops. If there is sudden rain and if the temperature is too low for the crops, the system will turn on the heat lamp due to which the temperature will be maintained. If due to high temperature, the moisture of the soils gets reduced, the system will turn on the water pump to maintain the moisture. Some crops need specific amount of light. If there is a situation that the surrounding light is low for the crops, then the system will turn on the light bulb to maintain the light intensity. In all these situations the farmer need not to run behind things like turning on water pump, light bulb, fan, etc. This system will take care of the maintenance of agriculture. All this data will be transmitted to the mobile device of the farmer.

II)BY USING ONLY SOIL MOISTURE SENSOR USING MICROCONTROLLER:

- https://www.researchgate.net/publication/373316667_DESIGN_OF_FERTIGATION_SYSTEM_CONTROL_IN_GREEN_HOUSE_BASED_ON_INTERNET_OF_THING_IoT

- Green house must be able to control the environments that are suitable for plant growth. However, manual watering must always be done at any time, which is time-consuming for farmers. Greenhouses with modern technology create automatic controls such as plant sprinklers. Thus, the time spent on watering plants is less than the manual system. In addition, farmers can save water which has been wasted all this time because they do not know the condition of water requirements for plants. With the development of the internet almost all over the world, giving changes to daily human activities. Internet of Things (IoT) technology allows objects to connect and communicate with each other. In this fertigation control device, IOT connects sensor devices and built with the ESP8266 module. The hardware design uses a microcontroller as a control method. The conclusion of this tool is that the system can do watering automatically. The system can do watering automatically, So that the condition of the plants can be maintained properly. Can display the status of soil moisture, conditions on the LCD and the Blynk applications.

CHAPTER 3

EXISTING METHOD

3.1 INTRODUCTION

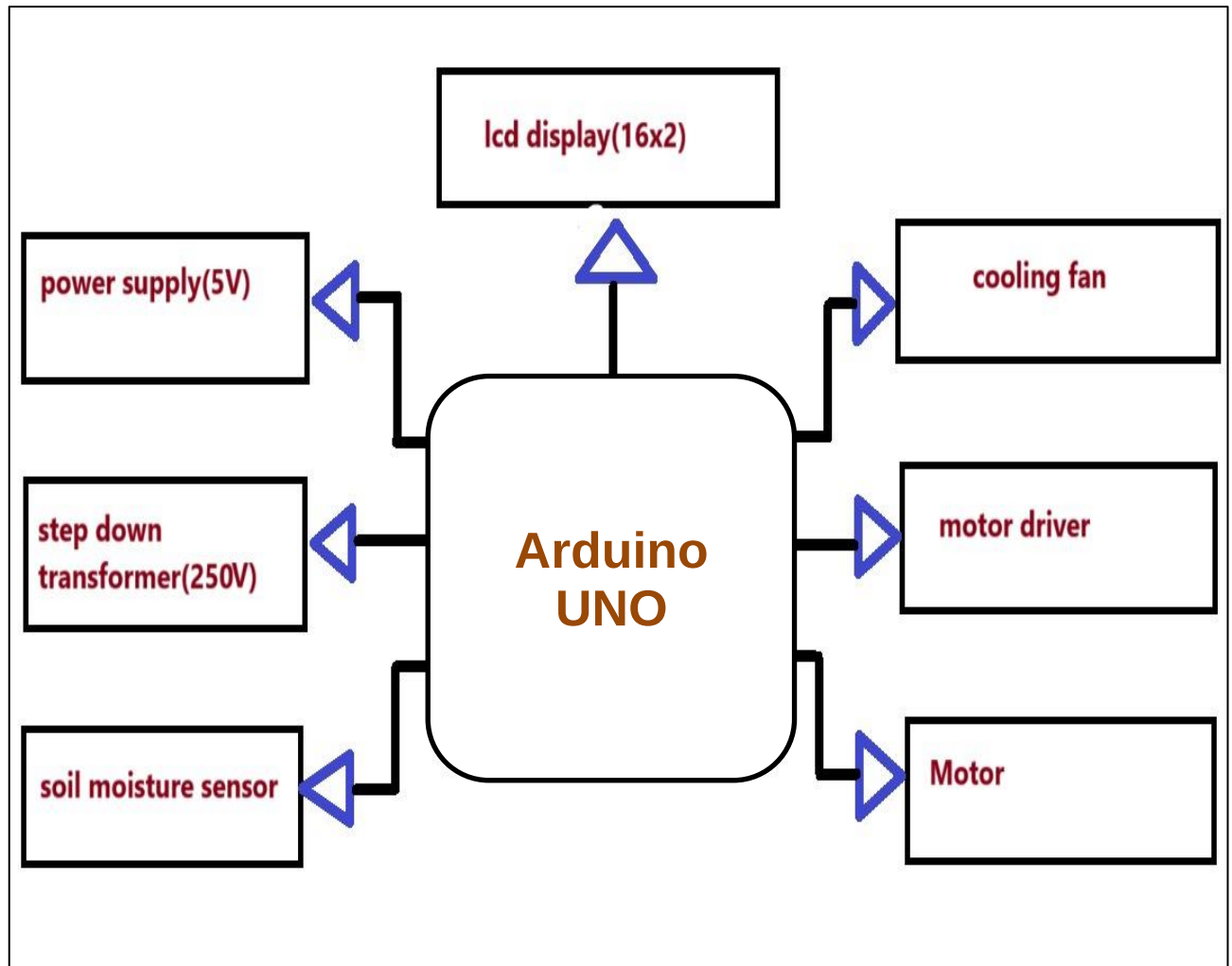


Fig 1: Block Diagram of Existing Method

Soil Moisture Sensor Plays major role when it occurs sudden changes in weather are not favorable for agriculture. Unpredicted rain can destroy the crops. Or less rain can dry off the crops. High temperatures are not good for crops. To prevent all these things, the farmers need to turn on and off the water pump, etc. Such situations can be difficult for farmers. This system is developed to ease the work of farmers. The farmer need not run behind all these things. These sensors are normally used to check volumetric water content,

and another group of sensors calculates a new property of moisture within soils named water potential. Generally, these sensors are named as soil water potential sensors which include gypsum blocks and tensiometer.

The soil moisture sensor is one kind of sensor used to gauge the volumetric content of water within the soil. As the straight gravimetric dimension of soil moisture needs eliminating, drying, as well as sample weighting. These sensors measure the volumetric water content not directly with the help of some other rules of soil like dielectric constant, electrical resistance, otherwise interaction with neutrons, and replacement of the moisture content. The relation among the calculated property as well as moisture of soil should be adjusted & may change based on ecological factors like temperature, type of soil, otherwise electric conductivity.

3.2 Sensors

DHT 11 Sensor:

DHT11 Temperature & Humidity Sensor features a temperature & humidity sensor complex with a calibrated digital signal output. By using the exclusive digital signal-acquisition technique and temperature & humidity sensing technology, it ensures high reliability and excellent long-term stability. This sensor includes a resistive-type humidity measurement component and an NTC temperature measurement component and connects to a high performance 8-bit microcontroller, offering excellent quality, fast response, anti-interference ability and cost-effectiveness.

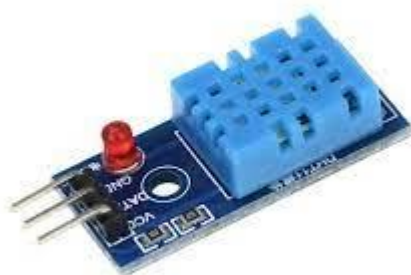


Fig 1.1: Working of DHT 11 SENSOR

Each DHT11 element is strictly calibrated in the laboratory and is extremely accurate on humidity calibration. The calibration coefficients are stored as

programmes in the OTP memory, which are used by the sensor's internal signal detecting process. The single-wire serial interface makes system integration quick and easy. Its small size, low power consumption and up-to-20 meter signal transmission making it the best choice for various applications, including those most demanding ones. The component is a 4-pin single row pin package. It is convenient to connect and special packages can be provided according to users request.

Communication with Microcontroller

- DHT11 uses only one wire for communication. The voltage levels with certain time value defines the logic one or logic zero on this pin.
- The communication process is divided in three steps, first is to send request to DHT11 sensor then sensor will send response pulse and then it starts sending data of total 40 bits to the microcontroller.
- This has DHT11 Send Response Signal and also Pullup and wait for response along with Microcontroller send Start Pulse.

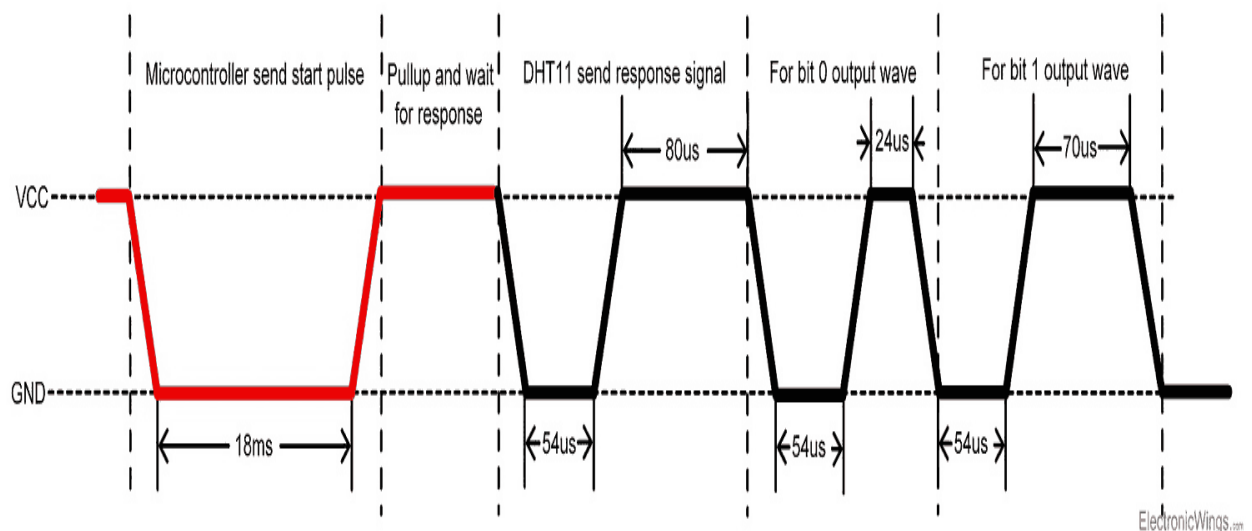


Fig1.2: Communication process of microcontroller

Start pulse (Request)

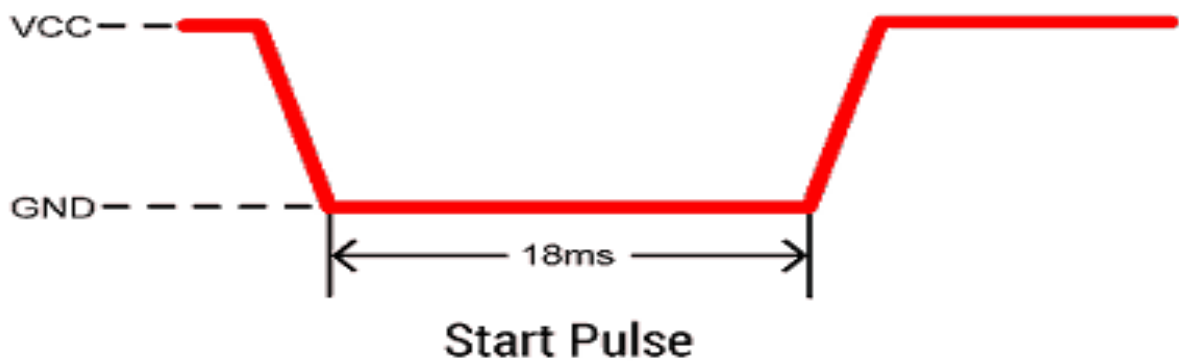


Fig 1.3:Request curve of microcontroller

- To start communication with DHT11, first we should send the start pulse to the DHT11 sensor.
- To provide start pulse, pull down (low) the data pin minimum 18ms and then pull up, as shown in diag.

Response

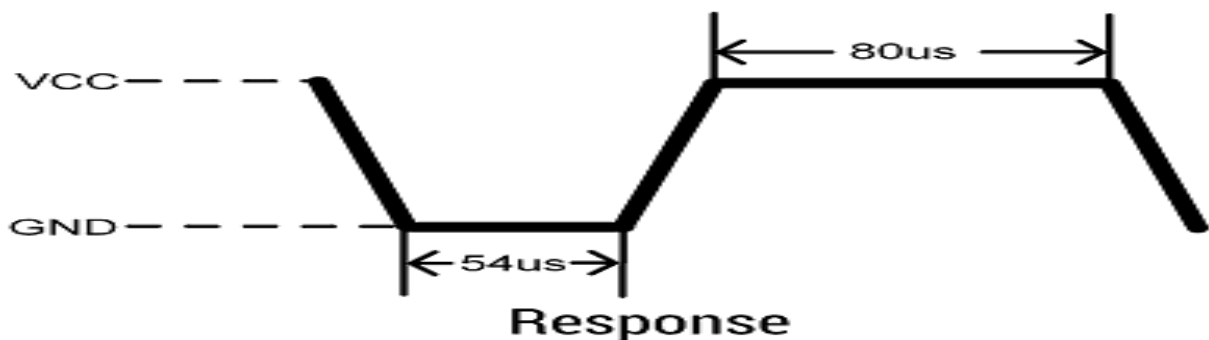


Fig 1.3:Response curve of micro controller

- After getting start pulse from, DHT11 sensor sends the response pulse which indicates that DHT11 received start pulse.
- The response pulse is low for 54us and then goes high for 80us.

Data

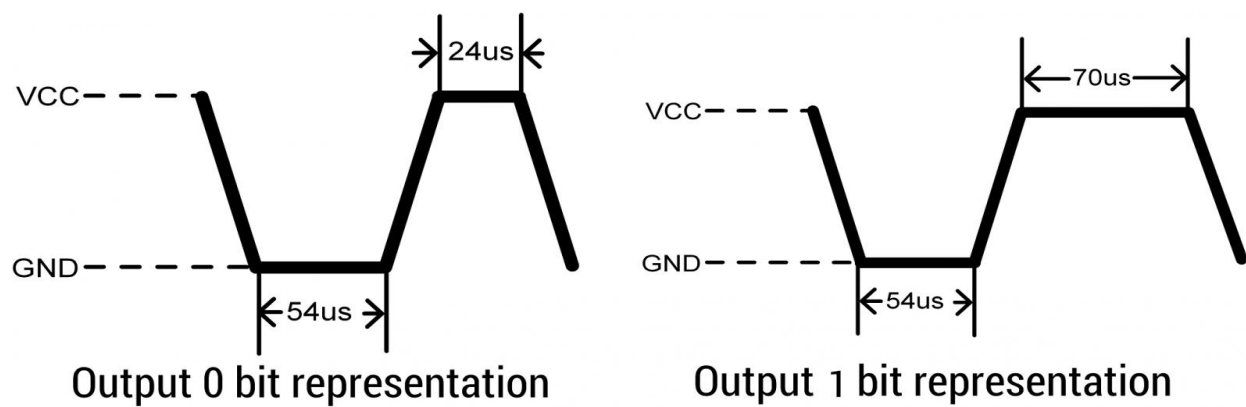


Fig 2.1:I)Data Bit Representation of Outputs

- After sending the response pulse, DHT11 sensor sends the data, which contains humidity and temperature value along with checksum.
- The data frame is of total 40 bits long, it contains 5 segments (byte) and each segment is 8-bit long.
- In these 5 segments, first two segments contain humidity value in decimal integer form. This value gives us Relative Percentage Humidity. First 8-bits are integer part and next 8 bits are fractional part.
- Next two segments contain temperature value in decimal integer form. This value gives us temperature in Celsius form.
- Last segment is the checksum which holds checksum of first four segments.
- Here checksum byte is direct addition of humidity and temperature value. And we can verify it, whether it is same as checksum value or not. If it is not equal, then there is some error in the received data.
- Once data received, DHT11 pin goes in low power consumption mode till next start pulse.
- And also shown Above with response pulse having output 0,1-bit Representation graphs with Vcc,GND.

End of frame

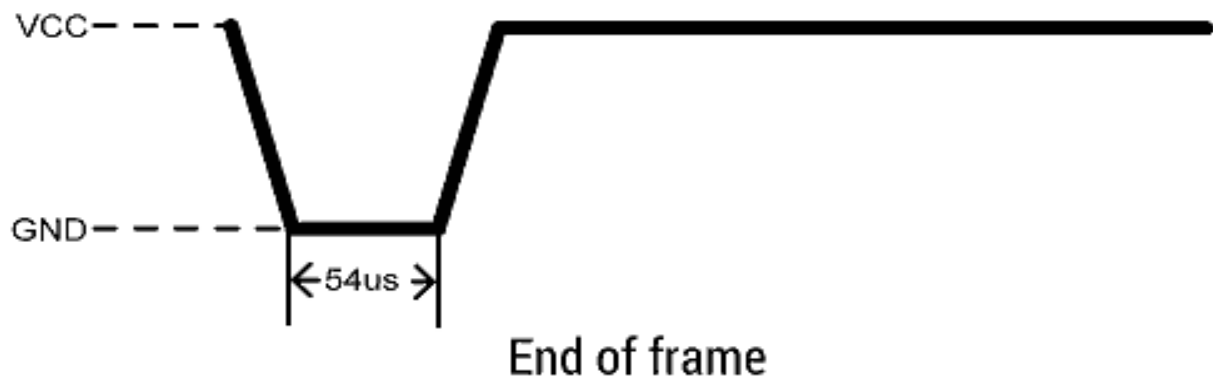


Fig 2.1:II)End of frame

- After sending 40-bit data, DHT11 sensor sends 54us low level and then goes high. After this DHT11 goes in sleep mode.

Applications:

It is applied in various fields such as weather monitoring, environmental analysis, and home automation systems.

Range:

Temperature range = 0°C to 50°C

Humidity range = 20% to 90% RH

Advantages:

- Low power consumption
- Relative humidity and temperature measurement
- Compact in size

Disadvantages:

- Low accuracy
- Limited operating range
- Fragile

LDR Sensor:

LDR is an acronym for Light Dependent Resistor. LDRs are tiny light-sensing devices also known as photoresistors. An LDR is a resistor whose resistance changes as the amount of light falling on it changes. The resistance of the LDR decreases with an increase in light intensity, and vice-versa .

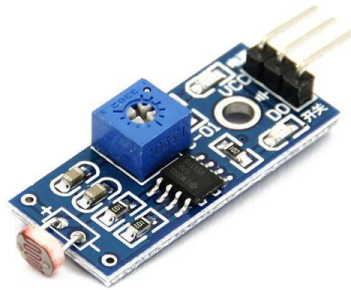


Fig 3.1:Working Principle of LDR

Let's delve into the working principle of the LDR Sensor, a key component in modern electronics. The LDR sensor, short for Light Dependent Resistor, operates on a fascinating principle that is both simple and effective. Light hitting an LDR module significantly alters its resistance, a key aspect of its operational principle.

The core of the LDR sensor working principle lies in photoconductivity. In the darkness, the LDR sensor exhibits high resistance, often in the range of megaohms. However, upon exposure to light, the electrons in the sensor's material are energized from the valence band to the conduction band. This transition reduces the resistance dramatically, often down to a few hundred ohms in bright light. This swift change in resistance in response to light intensity is what makes the LDR sensor working so reliable and useful in various applications. The working of the LDR sensor is not just about the change in resistance; it's also about how quickly this change occurs, making the LDR module highly responsive to light changes.

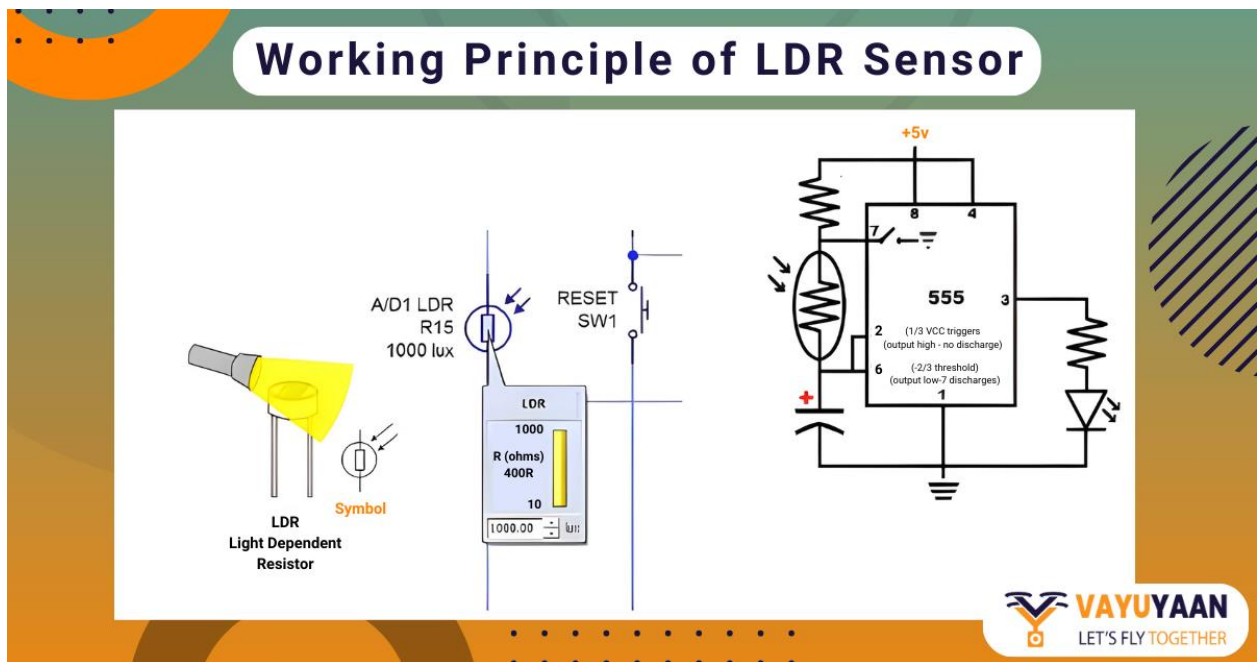


Fig 3.1 :Working Principle of LDR Sensor and various components

The wavelength of the incident light influences the working principle of the LDR sensor. This allows for the tailoring of the sensor's sensitivity for specific applications, thereby enhancing its versatility.

In essence, understanding the LDR sensor working principle offers insights into a device that is crucial in numerous applications, from automatic lighting to security systems. The LDR sensor excels in simplicity, responsiveness, and adaptability, proving essential in light-sensing technology.

By grasping the workings of the LDR sensor and its working principle, we unlock a deeper appreciation for this ingenious device, a testament to the innovative use of material properties in technology.

Variations in Resistance to Change in Light Intensity

The LDR Sensor's response to light is not just a simple on-off affair. It is more nuanced, with the resistance varying significantly based on the intensity of the light. This is where the LDR Sensor truly shines. In bright light, its resistance plummets, allowing more current to flow through. Conversely, in the dark, the resistance soars, reducing the current flow. The LDR Sensor's versatility and value lie in its precise control response to varying light intensities.

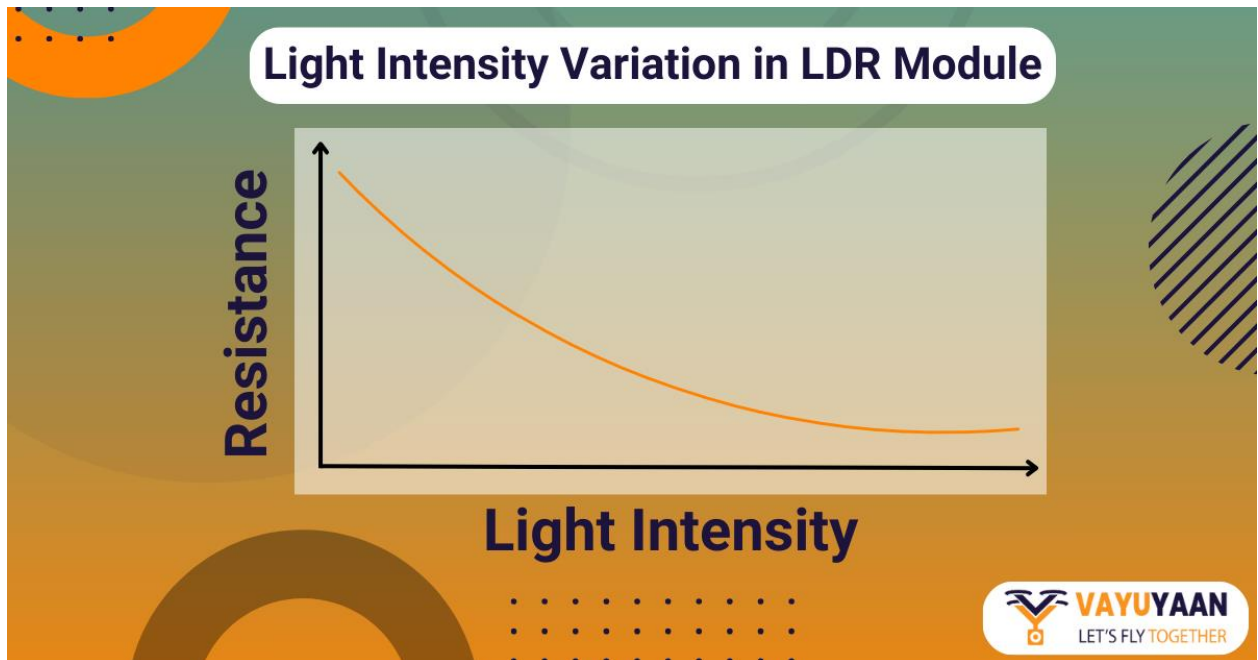


Fig 3.2:Light intensity curve of LDR Module

The LDR module does not just react to light; It is sensitive to the type of light, too. Different wavelengths of light can cause different responses in the LDR Sensor. The LDR Sensor's sensitivity to light's wavelength adds another layer of functionality, enabling it to be fine-tuned for specific applications.

Applications:

- Used in automatic lights that switch on and off according to light
- Laser-based security systems
- Solar Street Lamps
- Camera light meters

Advantages:

- High Sensitivity
- Simple and Small
- Low Cost
- Ease of Use

Disadvantages:

- Limited Operating Range
- Environmental Vulnerability
- Aging and Sensitivity Decay

Soil Moisture Sensor:

Soil moisture sensors are important tools for measuring soil moisture content, providing important data for a variety of applications, especially in agriculture. These sensors are based on the principle of change in ground capacitance or resistance.

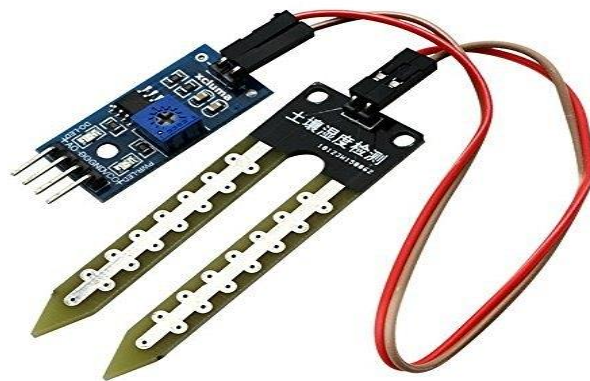
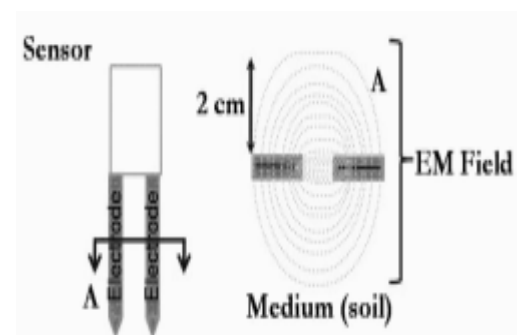


Fig 3.3:Working principle of Soil Moisture Sensor

How the Sensor Works

The Soil Moisture Sensor uses capacitance to measure dielectric permittivity of the surrounding medium. In soil, dielectric permittivity is a function of the water content. The sensor creates a voltage proportional to the dielectric permittivity, and therefore the water content of the soil.



The sensor averages the water content over the entire length of the sensor. There is a 2 cm zone of influence with respect to the flat surface of the sensor, but it has little or no sensitivity at the extreme edges. The figure above shows the electromagnetic field lines along a cross-section of the sensor, illustrating the 2 cm zone of influence.

Applications:

- Irrigation optimization: Precisely measuring soil moisture levels allows farmers to irrigate fields only when needed, preventing both overwatering and underwatering. This leads to efficient water usage, reduced pumping costs, and improved crop yields.

Advantages:

- Precise irrigation
- Prevents overwatering and underwatering

Disadvantages:

- Relatively slow response time to soil water changes.
- Less accurate in sandy soils.
- Sensitive to temperature and salinity

3.3 Motors:

Motors are electromechanical devices that convert electrical energy into mechanical motion. They play a vital role in a wide range of applications, from powering industrial machinery and vehicles to controlling household appliances and electronics. Motors come in various types, including DC (direct current) motors, AC (alternating current) motors, and specialized varieties like stepper and servo motors. Each type has unique characteristics suited to specific tasks, with factors like efficiency, torque, and speed varying between them. Motors are essential components in modern technology, driving everything from fans and pumps to robots and electric vehicles, making them indispensable in our daily lives and industrial processes.

Motor drivers:

A motor driver module is an electronic component that serves as an interface between microcontrollers or control systems and electric motors, enabling precise control over motor direction, speed, and protection through features like H-bridges, PWM control, and current limiting.



Fig 4.1:Motor Driver

Motors required:

1. Water Pump
2. Circulation Fans

3.3.1 Water pump:

Water pumps are mechanical devices that move water from one place to another. They are commonly used in various applications, such as supplying water to households, irrigating fields, and circulating coolant in engines. Water pumps play a critical role in ensuring the efficient and reliable distribution of water for domestic, agricultural, and industrial puposes.



Fig 4.2:Water Pump

3.3.2 Circulation fans:

Circulation fans are essential devices designed to improve air movement within enclosed spaces. They help distribute air evenly, preventing stagnant pockets and maintaining consistent temperatures. Commonly used in homes, offices, and industrial settings, circulation fans promote better ventilation, cooling, and heating efficiency. They are an economical and energy-efficient solution for enhancing indoor air quality and comfort.



Fig 4.3:Circulation Fans

CHAPTER 4

PROPOSED SYSTEM

4.1 INTRODUCTION

Below shown is the block diagram representation of the “Greenhouse Monitoring System Using IOT”

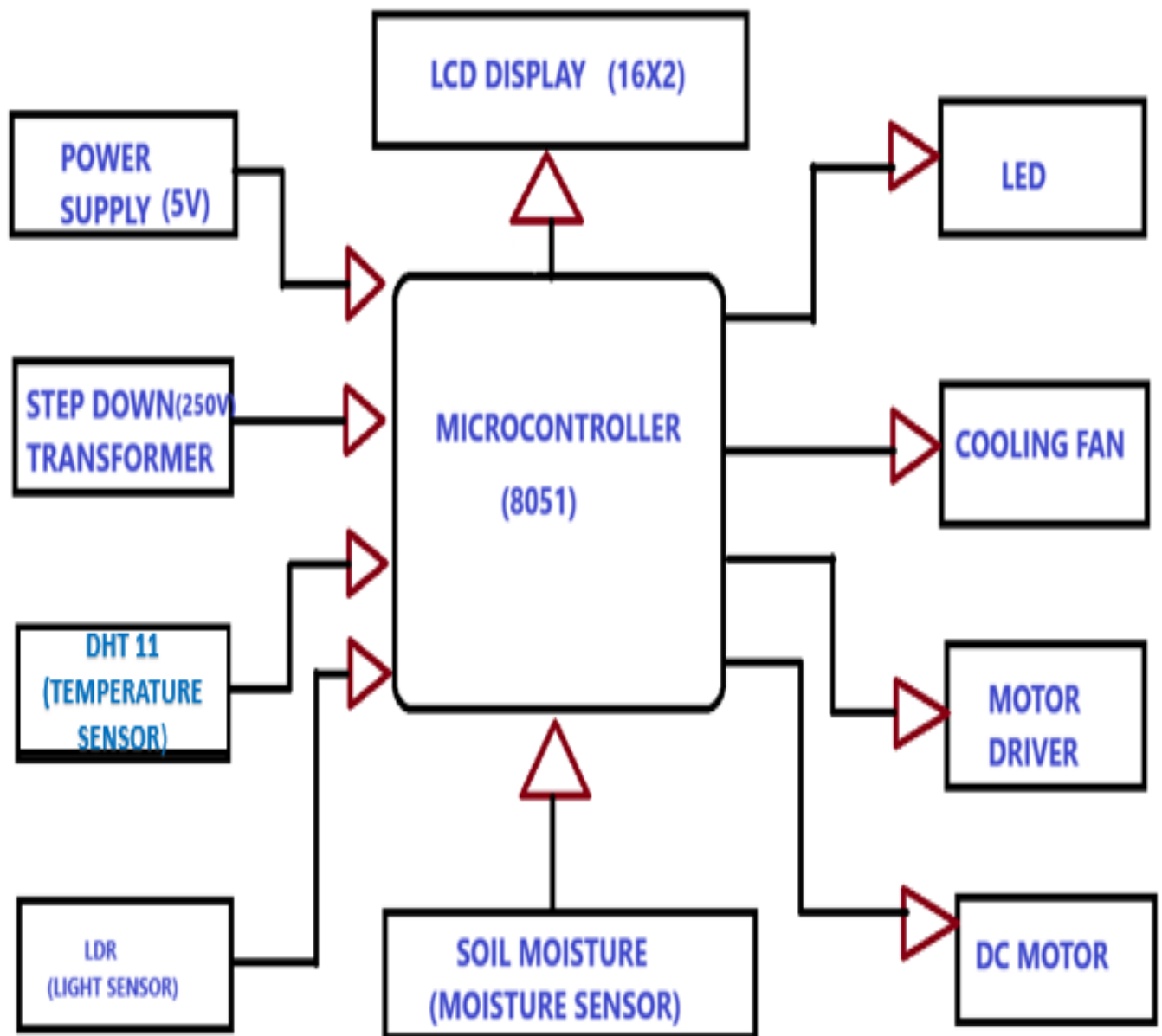


Fig 5.1:Block Diagram of Proposed System

4.2 Expanded Sensor Network:

The integration of IoT technology in greenhouse monitoring systems has led to the expansion of sensor networks, particularly with the inclusion of light and temperature sensors. These additional sensors play a crucial role in enhancing control over the greenhouse environment by providing real-time data on light intensity and temperature fluctuations. By complementing existing soil moisture sensors, growers can now access comprehensive environmental data necessary for informed decision-making. This expanded sensor network not only allows for a more detailed understanding of the greenhouse's microclimate but also enables growers to implement precise adjustments to optimize growing conditions. Whether it's adjusting light levels to match plant requirements or fine-tuning temperature settings, the integrated sensor network empowers growers to create customized environments conducive to plant growth and productivity.

4.3 Comprehensive Data Insights:

Implementing an IoT-based system in greenhouse monitoring offers unparalleled insights into the environment, providing growers with comprehensive data crucial for effective decision-making. Unlike traditional methods that offer limited information on specific parameters, IoT-based systems collect a wide range of data points from various sensors, including light, temperature, humidity, soil moisture, and more. This holistic approach to data collection and analysis allows growers to gain a deeper understanding of the greenhouse conditions, uncovering subtle trends and patterns that may go unnoticed with conventional monitoring methods. As a result, growers can make informed decisions regarding environmental control strategies, nutrient management, pest control, and irrigation scheduling, all of which contribute to optimized plant growth and increased productivity.

4.4 Smart Decision-Making:

Harnessing real-time data from multiple sensors empowers growers to make smart decisions in greenhouse management, leading to optimized plant growth and resource utilization. The integration of various sensors such as soil moisture, temperature, humidity, and light sensors provides a comprehensive view of the greenhouse

environment. This real-time data collection and analysis enable growers to monitor key parameters continuously, identify trends, and respond promptly to changing conditions. For example, by tracking soil moisture levels, growers can schedule irrigation more efficiently, ensuring plants receive adequate water without wastage or waterlogging, which can harm plant health.

4.5 Resource Optimization:

The implementation of IoT solutions in greenhouse management addresses various drawbacks in traditional methods, leading to significant reductions in inefficiencies. By leveraging IoT technology, growers can automate and optimize control systems, mitigating human errors and ensuring precise management of resources. This automation extends to critical areas such as irrigation scheduling, where IoT-enabled sensors can monitor soil moisture levels and autonomously adjust water flow to meet plant needs. Such precise control not only conserves water but also prevents overwatering, a common issue in traditional manual irrigation methods.

4.6 Enhanced Crop Management:

The IoT-based approach significantly outperforms traditional greenhouse methods by offering a more proactive and responsive management system. With real-time insights into crop health and environmental conditions, growers can detect issues early, such as pest infestations or nutrient deficiencies, and take immediate corrective actions. This proactive management not only improves crop yield and quality but also promotes sustainability by reducing resource wastage and environmental impact. By optimizing conditions for better overall plant growth, IoT-based crop management systems play a vital role in ensuring food security and supporting sustainable agriculture practices for future generations.

4.7 Proposed Greenhouse Monitoring System with Additional Sensors:

4.7.1 LDR Sensor (Light Dependent Resistor):

Measuring ambient light levels inside the greenhouse is essential for optimizing plant growth and productivity. By deploying sensors to monitor light levels,

growers can gather accurate data on natural light conditions within the greenhouse environment. This data not only helps in understanding the intensity of sunlight but also provides insights into daylight duration, crucial for determining photoperiod-sensitive plant behaviours such as flowering and fruiting. By analysing this information, growers can adjust artificial lighting schedules or shading systems to supplement natural light effectively, ensuring plants receive the optimal amount of light for their growth stages. This enhanced understanding of ambient light conditions plays a vital role in greenhouse management, enabling growers to create customized environments that promote healthy plant development, maximize yields, and improve overall crop quality.

4.7.2 DHT 11 Sensor (Temperature and Humidity):

The real-time data provided by temperature and humidity sensors allows growers to implement timely interventions such as adjusting ventilation, heating, or cooling systems to regulate conditions within the greenhouse. This proactive approach not only promotes plant health but also minimizes energy waste associated with excessive heating or cooling. Furthermore, by preventing extreme temperature or humidity fluctuations, growers can reduce the risk of crop losses due to environmental stressors, ensuring a more consistent and reliable supply of high-quality produce. Overall, the monitoring and management of temperature and humidity levels play a crucial role in modern greenhouse operations, supporting sustainable practices and optimizing plant growth outcomes.

4.7.3 Integration with Soil Moisture Sensor:

Comprehensive monitoring of soil moisture, light intensity, temperature, and humidity levels in the greenhouse provides growers with a holistic understanding of the environmental factors crucial for plant growth. By deploying sensors that continuously collect data on these parameters, growers gain valuable insights into the complex interactions within the greenhouse environment. This comprehensive approach to monitoring allows for a nuanced understanding of how variations in soil moisture, light availability, and climatic conditions impact plant health and development. Such detailed information empowers growers to make informed decisions regarding irrigation scheduling, lighting adjustments,

temperature control, and humidity management, all tailored to optimize conditions for precise and efficient agriculture practices.

The holistic data obtained from these sensors not only aids in monitoring individual parameters but also reveals interdependencies and correlations among different environmental factors. This integrated approach enhances decision-making processes for precision agriculture in greenhouse settings, enabling growers to implement targeted interventions that promote optimal plant growth while conserving resources. By leveraging real-time data analytics and smart control systems, growers can achieve higher crop yields, improve crop quality, and minimize environmental impacts associated with excessive resource usage. Ultimately, the combination of comprehensive monitoring and improved decision-making capabilities contributes to sustainable and productive agriculture practices, ensuring long-term success for greenhouse operations.

4.8 Advantages of the Proposed Method:

Fine-tuned environmental control based on multiple parameters is essential for optimizing crop growth and quality in greenhouse settings. By integrating sensors that monitor various factors such as soil moisture, light exposure, temperature, and humidity levels, growers can achieve precise management of environmental conditions. This data-driven approach allows for dynamic adjustments in irrigation schedules, lighting configurations, and climate control systems, ensuring that plants receive the optimal conditions for their growth stages. By fine-tuning these parameters, growers can minimize resource wastage while maximizing plant productivity, leading to increased crop yield and improved overall quality.

Optimizing irrigation practices, adjusting light exposure levels, and maintaining optimal climate conditions are key aspects of achieving successful crop management in greenhouses. By leveraging real-time data from multiple sensors, growers can detect subtle changes in environmental conditions and respond promptly with targeted interventions. This proactive approach not only prevents stressors such as water scarcity, inadequate lighting, or extreme temperature fluctuations but also fosters healthy plant development throughout the growth cycle. As a result, growers can achieve higher yields of premium-quality crops, meet market

demands consistently, and ensure sustainable agricultural practices that prioritize resource efficiency and environmental stewardship.

4.9 Overcoming Drawbacks of Existing Method:

The proposed greenhouse monitoring system seeks to address several limitations of the current method by integrating additional sensors such as LDR (Light Dependent Resistor) and DTH11 (Temperature and Humidity) sensors alongside the existing soil moisture sensor. One of the key challenges the system aims to overcome is the lack of insights into light conditions, which are critical for plant growth. By incorporating an LDR sensor, growers can now accurately measure light intensity levels within the greenhouse, allowing for better optimization of lighting schedules and ensuring plants receive adequate light for photosynthesis and growth.

Furthermore, the inclusion of DTH11 sensors enhances environmental monitoring by providing real-time data on temperature and humidity levels. This comprehensive approach not only helps in maintaining optimal climate conditions but also allows growers to detect and mitigate potential issues such as temperature extremes or high humidity, which can negatively impact plant health. By leveraging data from multiple sensors, the system enables growers to make informed decisions regarding irrigation, ventilation, and lighting adjustments, leading to more efficient resource utilization and improved crop productivity. Overall, the integration of LDR and DTH11 sensors with the existing soil moisture sensor represents a significant step towards achieving a more holistic and data-driven approach to greenhouse environmental management, benefiting both growers and the overall sustainability of agricultural practices.

CHAPTER 5

SOFTWARE REQUIREMENTS

```
#include<reg51.h>
#include<stdio.h>
#include<string.h>
#include <stdlib.h>

sbit rs=P2^6;// lcd register sellect
sbit en=P2^7;// lcd enable pin

sbit soil=P2^5;
sbit motor1=P3^2;
sbit motor2=P3^3;
sbit ldr=P2^0;
sbit led=P2^1;
sbit fan1=P2^3;
sbit fan2=P2^4;
sbit DHT11=P3^7;          /* Connect DHT11 output Pin to P2.1 Pin */
int I_RH,D_RH,I_Temp,D_Temp,Checksum;

void lcdcmd(char a);
void lcddata(char a);
void delay(int a);
void printlcd(char *a); // user defined functions
void timer_delay20ms()    /* Timer0 delay function */

{
TMOD = 0x01;
TH0 = 0xB8;          /* Load higher 8-bit in TH0 */
TL0 = 0x0C;          /* Load lower 8-bit in TL0 */
```

```

TR0 = 1;          /* Start timer0 */
while(TF0 == 0);  /* Wait until timer0 flag set */
TR0 = 0;          /* Stop timer0 */
TF0 = 0;          /* Clear timer0 flag */}void timer_delay30us()      /* Timer0 delay
function */
{
TMOD = 0x01;      /* Timer0 mode1 (16-bit timer mode) */
TH0 = 0xFF;       /* Load higher 8-bit in TH0 */
TL0 = 0xF1;       /* Load lower 8-bit in TL0 */
TR0 = 1;          /* Start timer0 */
while(TF0 == 0);  /* Wait until timer0 flag set */
TR0 = 0;          /* Stop timer0 */
TF0 = 0;          /* Clear timer0 flag */
}

void Request()     /* Microcontroller send request */
{
DHT11 = 0;        /* set to low pin */
timer_delay20ms(); /* wait for 20ms */
DHT11 = 1;        /* set to high pin */
}

void Response()    /* Receive response from DHT11 */
{
while(DHT11==1);
while(DHT11==0);
while(DHT11==1);
}

int Receive_data() /* Receive data */
{

```

```

int q,c=0;
for (q=0; q<8; q++)
{
while(DHT11==0);/* check received bit 0 or 1 */
timer_delay30us();
if(DHT11 == 1)    /* If high pulse is greater than 30ms */
c = (c<<1)|(0x01);/* Then its logic HIGH */
else             /* otherwise its logic LOW */
c = (c<<1);
while(DHT11==1);}
return c;
}

void main()
{
int d1,d2;

//unsigned char dat[20];

lcdcmd(0x38); //Select 16x2 LCD
lcdcmd(0x06); //Increment Cursor
lcdcmd(0x0E); //Display ON
lcdcmd(0x01);
lcdcmd(0x80);
printf("DHT11");
//lcdcmd(0x01);
delay(100);
while(1)
{
Request();    /* send start pulse */
Response(); /* receive response */
}
}

```

```

I_RH=Receive_data();    /* store first eight bit in I_RH */
D_RH=Receive_data();    /* store next eight bit in D_RH */
I_Temp=Receive_data();  /* store next eight bit in I_Temp */
D_Temp=Receive_data();  /* store next eight bit in D_Temp */
Checksum=Receive_data();/* store next eight bit in CheckSum */

if ((I_RH + D_RH + I_Temp + D_Temp) != CheckSum)
{
    printlcd("    Error    ");// print error in lcd
}
else
{
    /*char ir=I_RH;char dr=D_RH;char it=I_Temp;char dt=D_Temp;
    lcdcmd(0x01);
    lcdcmd(0x80);printlcd("Hm=");
    lcddata(ir);
    lcddata('.');
    lcddata(dr);
    printlcd("Tm=");
    lcddata(it);
    lcddata('.');
    lcddata(dt);*/

    TMOD=0x20;
    SCON=0x50;
    TH1=-3;
    TR1=1;
    d1=I_Temp/10;
    d2=I_Temp%10;
    lcdcmd(0x80);

```

```

printf("Tem:");
lcddata(d1+48);
lcddata(d2+48);
lcddata(0xDF);
lcddata('C');
lcdcmd(0x89);
printf("Hum:");
d1=I_RH/10;
d2=I_RH%10;
lcddata(d1+48);
lcddata(d2+48);
lcddata('%');
lcdcmd(0xc4);
delay(200);

```

```

if(I_Temp >20)
{
fan1=1;
fan2=0;
}
else
{
fan1=0;
fan2=0;
}
}
if (ldr == 1)
{
led=1;
lcdcmd(0xc0);

```

```

    printlcd("Light on");
}
else
{
    led=0;
    lcdcmd(0xc0);
    printlcd("Lignt of");
}
if(soil==1)
{
    motor1=1;
    motor2=0;
    lcdcmd(0xc9);
    printlcd("soil dry");
}
else
{
    motor1=0;
    motor2=0;
    lcdcmd(0xc9);
    printlcd("soil wet");
}
delay(500);
}
}

void lcdcmd(char a)
{
    P1=a;
    rs=0;
    en=1;

```

```

        delay(10);
        en=0;
    }
void lcddata(char a)
{
    P1=a;
    rs=1;
    en=1;
    delay(10);
    en=0;
}
void printlcd(char *a)
{
    while(*a)
        lcddata(*a++);
}
void delay(int a)
{
    int i,j;
    for(i=0;i<a;i++)
        for(j=0;j<1275;j++);
}

```

SOFTWARE TOOLS

5.1 Keil u Version:

Keil C51 is an integrated development environment (IDE) specifically designed for 8051 microcontrollers, providing developers with a user-friendly interface, a highly optimized C compiler, a powerful debugger, and simulation tools for peripheral devices. Its comprehensive code libraries tailored for 8051 microcontrollers, along with support for real-time operating systems and compatibility with version control systems, streamline the development and debugging processes for embedded applications. By offering a range of tools and features geared towards 8051-based systems, Keil C51 simplifies the creation of reliable and resource-efficient software, making it an invaluable asset for embedded system developers seeking to optimize their development workflows and produce high-quality embedded applications.

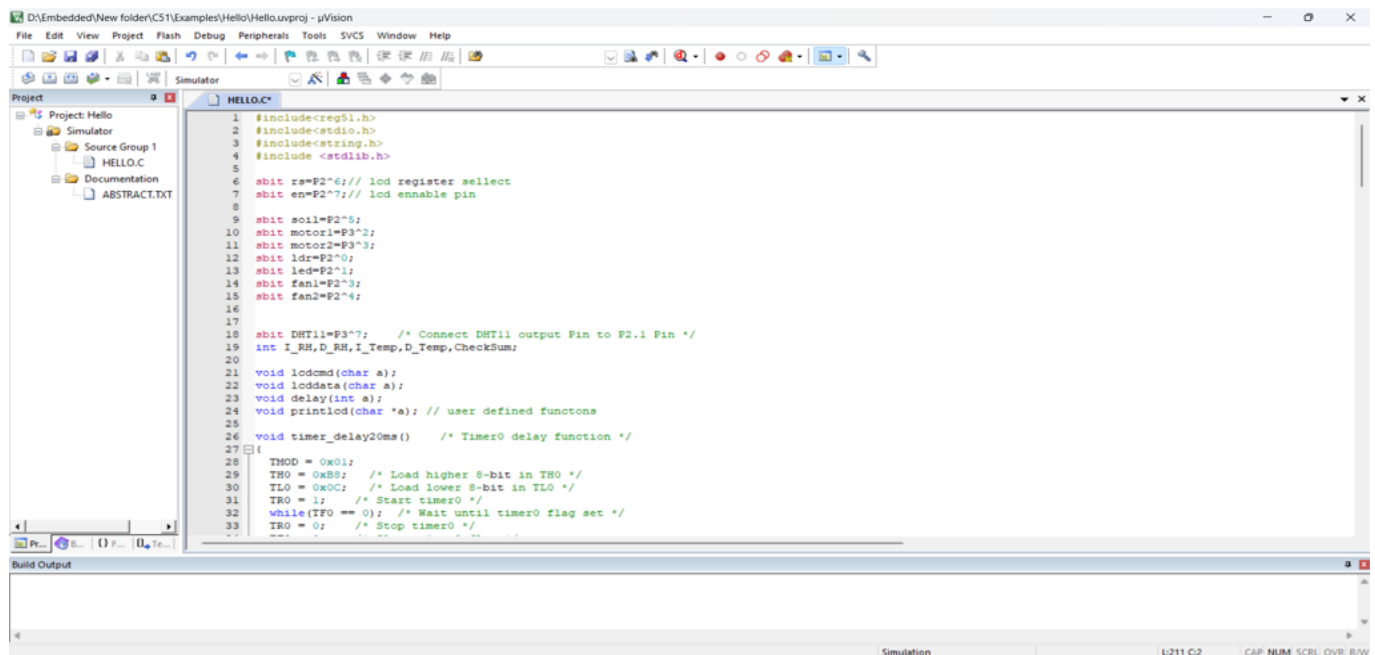


Fig 6.1:Code in Keil C51

5.2 Nuvoton:

The ISP-ICP utility software is a tool developed by Nuvoton Technology specifically designed for programming their microcontrollers and programmable devices in-circuit, offering a streamlined process that eliminates the necessity of physically removing the devices from printed circuit boards or systems. This utility facilitates in-circuit programming (ICP) and in-system programming (ISP), allowing developers and engineers to update firmware, configure settings, and perform debugging tasks directly on the target microcontroller or device without the inconvenience of removing it from its operational environment. By providing a convenient and efficient programming solution, the ISP-ICP utility software enhances productivity and reduces downtime in embedded system development and maintenance tasks involving Nuvoton microcontrollers and programmable devices.

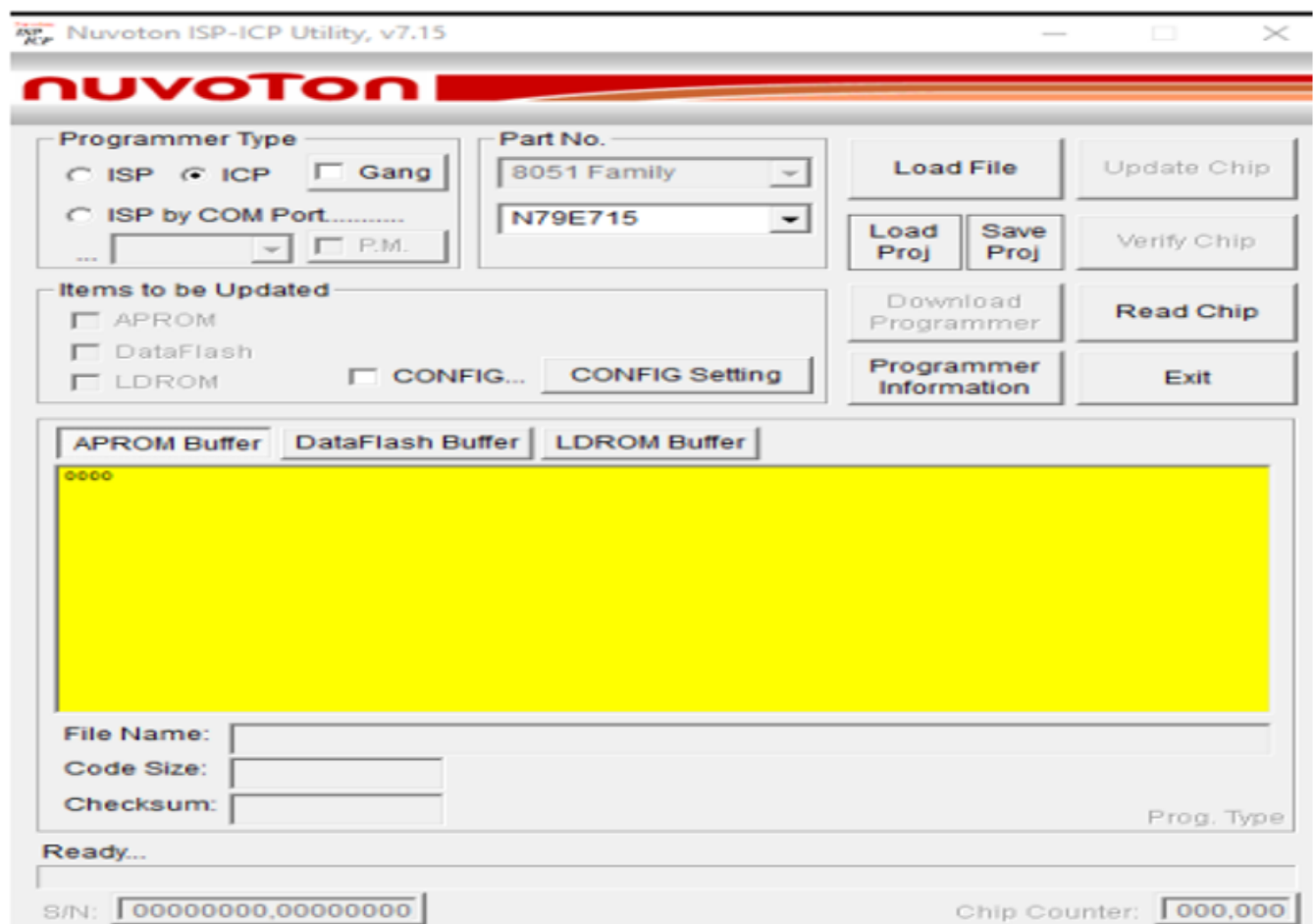


Fig 6.2:Nuvoton ISP-ICP Utility Software

5.3 Proteus:

Proteus stands out as a versatile and extensively used software tool for electronic circuit design, simulation, and PCB layout creation. Its rich library of components and robust microcontroller support empowers engineers, hobbyists, and students to virtually design and test electronic circuits and systems before physical implementation. The software's interactive simulation and debugging features, complemented by 3D visualization and mixed-signal support, offer a comprehensive platform for accurately modeling and validating electronic designs. Furthermore, Proteus plays a crucial role in PCB layout design by facilitating a seamless transition from schematic design to physical board layout, streamlining the entire design process. Its ability to simulate real-world conditions and interactions, along with its support for various microcontrollers and components, makes it an invaluable tool for electronic design projects of varying complexities and scales.

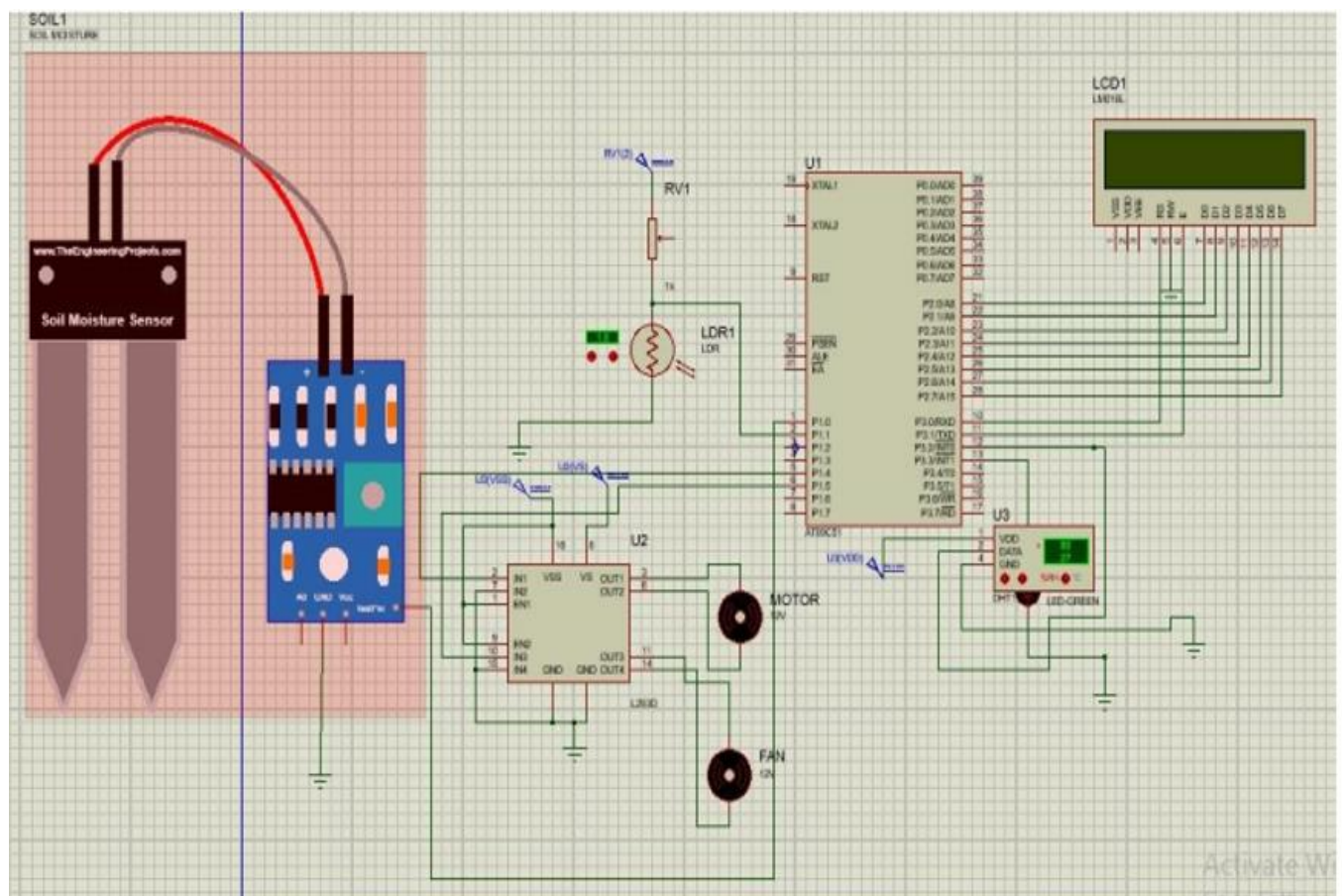


Fig 6.3:Circuit Diagram in Proteus Software

CHAPTER 6

RESULT

The soil moisture sensor checks the level of moisture in the soil and obtained readings in LCD 16X2 Display, By comparing the obtained readings with the threshold level the output is given as binary output which is '1' or '0', These readings are sent to the microcontroller. If these readings of moisture are very low, then the microcontroller turns on the motor which pumps water and makes the soil more hydrated. The DHT sensor monitors the temperature which surrounds the plants. The sensor gives a 16-bit digital data as output. In the 16-bit of digital data, 8-bits of data represents temperature and other 8-bits of data represents the humidity. These readings are sent to the microcontroller. The temperature which is suitable for the plant growth is set as the threshold temperature. If the temperature around the plants are not suitable for the growth of plants, which means if the temperature is greater than the threshold temperature, then the microcontroller activates the fans to maintain the temperature and humidity suitable for healthy growth of plants.

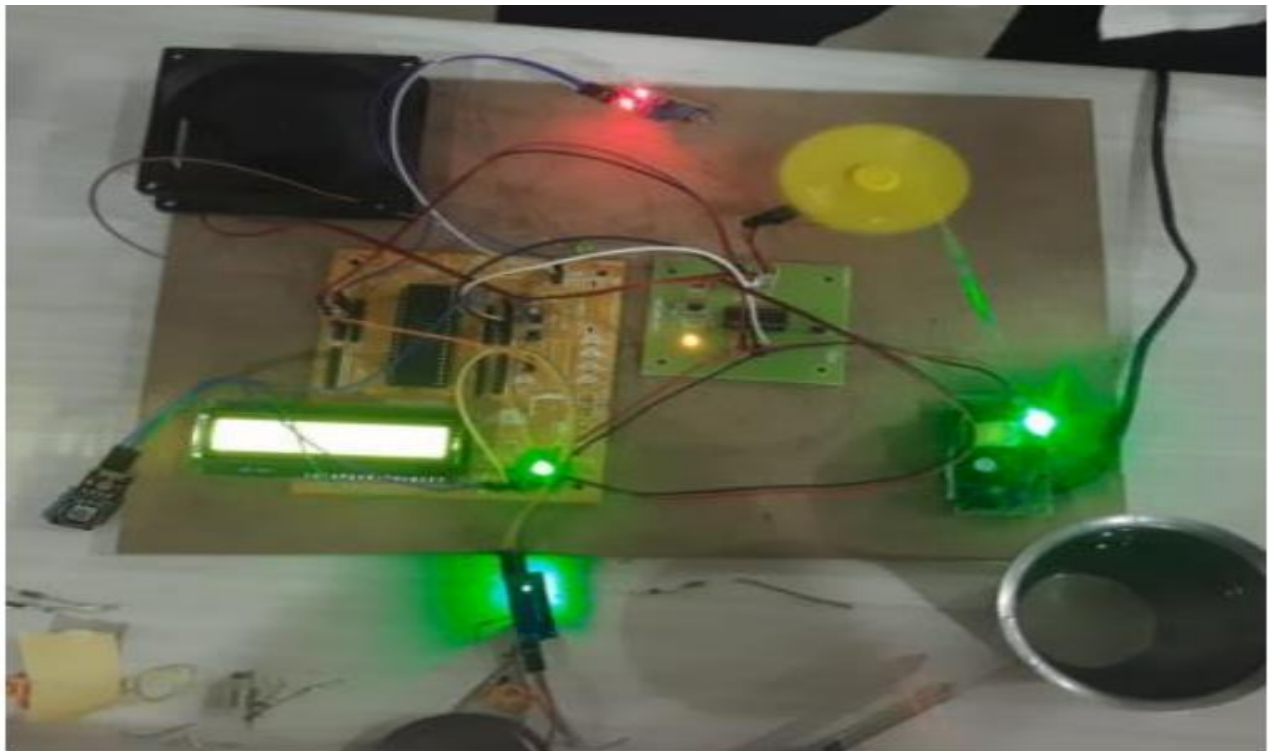


Fig 7.1 : Working Of Sensors In Micro Controller Board

The LDR sensor analyses the intensity of light around the plants. LDR sensor gives analog data as output which ranges between 0-255. Same as the previous sensors a threshold intensity of light is set. The threshold level is set based on the level at which the amount of light is sufficient for photosynthesis of the plant. If the intensity of light falls below the threshold level then the microcontroller turns on supplementary artificial lighting which results in optimal photosynthesis. The LCD displays the temperature and humidity values ,LED on or off, soil conditions as wet or dry state with the temperature and humidity values as shown below in LCD Display.



Fig 7.2:Sensor Values in LCD 16X2

CHAPTER 7

CONCLUSION

In conclusion, the integration of 8051 microcontrollers, temperature sensors, soil moisture sensors, LEDs, and motors into a green house monitoring system presents a powerful and sophisticated tool for optimizing plant growth and enhancing overall greenhouse Efficiency. The 8051 Microcontroller serves as the brain of the system, overseeing data acquisition, process and control. Its ability to efficiently manage multiple tasks makes it suitable choice for orchestrating the various components within the greenhouse environment, ensuring seamless operation and precise control. By incorporating temperature sensors the system enables precise monitoring and regulation of greenhouse temperatures, creating an optimal climate for plant growth. This capability is crucial for maintaining consistent environmental conditions, especially in regions with fluctuating weather patterns. The real-time data provided by the sensors allows for immediate adjustments to heating ,cooling and ventilation systems, ensuring that plants thrive in optimal conditions throughout their growth cycle without Human Effort.

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